

Laser Welding Parameter Optimization and Porosity Control

Abstract

This study investigates laser welding of Q355 structural steel and connects processing parameters to mechanical performance and porosity control.

Introduction

Laser welding quality depends on heat input, shielding gas and keyhole stability.

Materials and Methods

The base material was Q355 steel with a thickness of 3 mm.
The laser power was 2.4 kW and travel speed was 12 mm/s.

Table 1 Welding parameters

Group	Laser power	Speed
A	2.0 kW	10 mm/s
B	2.4 kW	12 mm/s
C	2.8 kW	14 mm/s

Results

The highest tensile strength was 612 MPa for parameter group B.

Discussion

Porosity formed because shielding gas was trapped during unstable keyhole collapse; stable keyhole behavior reduced pore formation.

Conclusions

A laser power of 2.4 kW produced the best combined mechanical performance in this experiment.

References

- [1] Zhang et al. Laser welding process monitoring. 2024.
- [2] Wang et al. Keyhole stability and pore formation. 2023.